
Alignment as a Linear Social Choice Problem

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Whenever an algorithm makes a decision that affects the welfare of more than
2 one person, it confronts a social choice problem. However, the standard approach
3 to aligning AI systems—reinforcement learning from human feedback (RLHF)—
4 aggregates feedback from heterogeneous individuals as if it were generated by a
5 single representative agent, which can produce systems that violate basic social
6 welfare criteria such as Pareto efficiency. Taking the inherently social nature of
7 alignment seriously requires that we do better. Fortunately, the fields of social
8 choice and mechanism design have spent decades developing a rich theoretical and
9 algorithmic toolkit for studying exactly this class of problem. We show how to
10 bridge these two worlds by first show that the standard preference model assumed
11 by RLHF and related alignment procedures is formally equivalent to a linear social
12 choice problem. We then apply this equivalence to [do useful stuff...] . We intro-
13 duce alignment externalities—the marginal effect of reweighting an individual’s
14 training data on downstream social welfare—and show that they admit an influence-
15 function characterization that geometrically explains when and why anonymous
16 RLHF distorts inferred preferences away from socially efficient outcomes. Our
17 analysis reveals fundamental recoverability constraints: many Pareto-efficient pref-
18 erence parameters cannot be implemented by anonymous protocols without altering
19 how queries—and thus influence—are distributed across individuals.

1 Introduction

- 21 • Aligning an algorithm that makes decisions impacting more than one person requires solving
22 a social choice problem, whether explicitly or implicitly.
- 23 • The standard approach this problem for AI algorithms, reinforcement learning from human
24 feedback (RLHF), has been shown to violate basic welfare properties by recent work
25 (CITES).
- 26 • As we deploy these algorithms in broader, more consequential social contexts, or start using
27 them to intermediate social and political interactions (e.g., contracting, voting, organizational
28 decision making), we need to do this well
- 29 • Here, we introduce a framework that shows how the AI social choice problem can be studied
30 using a powerful set of tools from economics (in the linear preference setting, which is
31 typically assumed)
- 32 • We first use the framework to fully characterize the geometry of RLHF distortion
- 33 • We then use the framework to fully characterize the set of strategy-proof AI alignment
34 protocols
- 35 • Finally, we use the framework to fully characterize a set of mechanisms that encode flexible
36 social objectives, such as public spiritedness and bounded harm

- We demonstrate the power of these theoretical results and further study their implications in multiple rich domains using real-world data

2 Related Work

1. Ge et al. [2024a]: “Axioms for alignment.” Critically, relative to this and other recent papers, we focus on the distribution of queries and the inter-relationships between them, not distortion on a single query.
2. Gözl et al. [2025]: Bounds distortion for an individual choice problem; studies connection between BTL and Borda count
3. Procaccia and Rosenschein [2006]: introduced distortion
4. Shirali et al. [2025]: Derives the second-order Taylor expansion + variance estimation; also explores individual-level reward modeling
5. Chakraborty et al. [2024]: Apply EM on latent preference variables
6. Chidambaram et al. [2024]: Cluster annotators and fit policies for each cluster via EM
7. Park et al. [2024]: Characterize RLHF convergence for a MARL setup
8. Siththaranjan et al. [2023]: Note bias of learned utility with DPO RLHF, advocate “distributional learning”
9. Halpern et al. [2025]
10. Munos et al. [2024]: Nash learning from human feedback
11. Ge et al. [2024b, 2025]: papers on linear utility functions; learning rates & distortion
12. Green and Laffont [1977]: theorem that VCG mechanisms are essentially unique as efficient DSIC mechanisms
13. Buening et al. [2025]: strategyproof RLHF
14. Alon et al. [2011]: (“sum of us”) strategyproof sampling
15. Min et al. [2025]: analyze RLHF via influence functions
16. Fein and Aranguiz-Dias [2025]: “Influence Functions for Preference Dataset Pruning”
17. Sun et al. [2025]: “Uncertainty and influence aware reward model refinement for reinforcement learning from human feedback”
18. Cao et al. [2025]: “SCAR: Shapley Credit Assignment for More Efficient RLHF”
19. Shen et al. [2024]: “Towards data-centric rlhf: Simple metrics for preference dataset comparison”
20. Conitzer et al. [2024]: “Position: Social choice should guide ai alignment in dealing with diverse human feedback”

3 Preliminaries

Let X be a set of contexts and Y a set of possible actions (e.g., choice problems and choices). There are N agents. Each agent n has preferences represented by a utility function $u_n : X \times Y \rightarrow \mathbb{R}$. For a given context and binary set of actions (x, y, y') , we assume the agent selects according to a stochastic choice function with a sigmoid link, as in the Bradley-Terry-Luce (BTL) model

$$p(y \succ y' \mid y, y', x) = \sigma(u_n(y \mid x) - u_n(y' \mid x)) \quad (1)$$

where $\sigma(a) = \frac{\exp(a)}{1 + \exp(a)}$ and $u_n : X \times Y \rightarrow \mathbb{R}$ describes n ’s decision utility from each option. Let $w \in \Delta_N$ be an arbitrary N -dimensional vector of social welfare weights. The utilitarian social welfare function is then defined as a weighted average of the individual utility generated by the choice

$$\bar{u}(y \mid x) = \sum_n w_n u_n(y \mid x) \quad (2)$$

Each context–action pair is embedded into a feature space $\phi(x, y) : X \times Y \rightarrow \mathbb{R}^k$ (e.g., the final activation layer of an LLM). The first important assumption is that each agent’s utility is linear in this feature space, i.e. can be represented as an inner product

$$u_n(y | x) = \phi(x, y)^\top \theta_n \quad (3)$$

for some latent preference parameter $\theta_n \in \mathbb{R}^k$.

A direct consequence of linearity is that the utilitarian social welfare function can be represented as a weighted average of the individual preference parameters. Specifically, if we define $\bar{\theta} = \sum_n w_n \theta_n$, then

$$\bar{u}(y | x) = \phi(x, y)^\top \bar{\theta} \quad (4)$$

For any stochastic choice rule that is a function of the utility differences (including the Logistic), linearity implies that each contextual choice problem is completely characterized by a vector

$$\alpha(x, y, y') = \phi(x, y) - \phi(x, y') \quad (5)$$

For notational simplicity, we will write $\alpha_z \in \mathbb{R}^k$ to denote the vector that corresponds to a specific choice problem $z \in Z = X \times Y \times Y$. For expositional simplicity, we assume the set of candidate queries Z is finite.

3.1 Social Welfare, Utilitarian Efficiency, and Pareto Efficiency

Let q_{dep} denote the deployment distribution over choices the model will make that affect the entire population. For each individual n , define deployment welfare as the utility induced by the model’s policy

$$W_n(\theta) = \mathbb{E}_{z \sim q_{\text{dep}}} \left[2 \alpha_z^\top \theta_n \left(\sigma(\alpha_z^\top \theta) - \frac{1}{2} \right) \right] \quad (6)$$

Utilitarian social welfare is then defined as the weighted average $W(\theta) = \sum_{n=1}^N w_n W_n(\theta)$, where $w \in \Delta_N$ is an arbitrary vector of social welfare weights.

The logistic choice function becomes less stochastic as a preference parameter is scaled up (or, equivalently, its temperature is reduced). This mechanically implies that simply scaling a parameter up can increase welfare, which requires that we slightly modified the standard approach to defining utilitarian efficiency.

Lemma 1. *Let a preference set $\{\theta_n\}_{n=1}^N$ and set of social welfare weights $w \in \Delta_N$ be given. We have*

$$\lim_{\beta \rightarrow \infty} W(\beta \bar{\theta}_w) = \sup_{\theta \in \mathbb{R}^k} W(\theta) \quad (7)$$

Hence, the utilitarian preference parameter $\bar{\theta}_w$ defines a ray that asymptotically attains the maximal achievable total welfare.

Let $C = \text{Conv}(\{\theta_n\}_{n=1}^N)$ denote the convex hull of the population’s preference parameters. Lemma 1 shows that every member of C is an asymptotically welfare-maximizing direction for some set of welfare weights and, conversely, that no parameter outside the set can asymptotically generate strictly more welfare than the best direction in C when scaled. We will therefore refer to $P = \text{cone}(C)$ as the set of “utilitarian efficient” parameters.

4 Alignment as a Social Choice Problem

A critical feature of the linear utility representation is that a deployed model’s impact on welfare can be completely characterized by a single vector.

Definition 2. The **impact** of a deployed model at parameter θ is fully represented by a vector

$$\psi(\theta) = \mathbb{E}_{z \sim q_{\text{dep}}} [2 \alpha_z (\sigma(\alpha_z^\top \theta) - \frac{1}{2})] \in \mathbb{R}^k \quad (8)$$

The impact vector is a sufficient statistic for how the model impacts welfare, both at the level of each individual and at the population level. Specifically, we can re-write the welfare of an agent n as

114 $W_n(\theta) = \theta_n^\top \psi(\theta)$ and of the entire population as $W(\theta) = \sum_{n=1}^N w_n \theta_n^\top \psi(\theta)$.¹ Taking an image of
 115 ψ over the space of all possible deployment parameters traces out a set of feasible impacts

$$\Psi = \{\psi(\theta) : \theta \in \mathbb{R}^k\} \subseteq \mathbb{R}^k \quad (9)$$

116 The following proposition characterizes key properties of the impact mapping and feasible impact set,
 117 which determines the possibilities frontier of individual and social welfare.

118 **Theorem 3.** *Let S_{dep} be the subspace of $\mathbb{R}^k$ spanned by the deployment queries. Then:*

- 119 1. **Convex Image:** $\Psi \upharpoonright_{S_{\text{dep}}}$ is a convex set.
- 120 2. **Zero Invariance:** Ψ contains the origin. In particular, $\psi(0) = 0$.
- 121 3. **Antisymmetry:** $\psi(\theta) = -\psi(-\theta)$ for all $\theta \in \mathbb{R}^k$.
- 122 4. **Boundedness:** $\sup_{\theta \in \mathbb{R}^k} \psi(\theta) \leq 2 \mathbb{E}_{z \sim q_{\text{dep}}} [\|\alpha_z\|]$.
- 123 5. **Smooth & Invertible:** ψ is a smooth C^∞ function from $\mathbb{R}^k$ onto Ψ and ψ^{-1} is a smooth
 124 C^∞ function from Ψ onto $\mathbb{R}^k$. Moreover, ψ^{-1} is a smooth C^∞ function from Ψ onto $\mathbb{R}^k$.
- 125 6. **Monotonic Scaling:** For any $\theta \in \mathbb{R}^k$ and $\beta \geq 0$, we have

$$\frac{\partial \psi(\beta \theta)}{\partial \beta} = \mathbb{E}_{z \sim q_{\text{dep}}} [2(\alpha_z^\top \theta) v(\beta \alpha_z^\top \theta) \alpha_z] \quad (10)$$

126 and, therefore,

$$\theta^\top \frac{\partial \psi(\beta \theta)}{\partial \beta} = \mathbb{E}_{z \sim q_{\text{dep}}} [2(\alpha_z^\top \theta)^2 v(\beta \alpha_z^\top \theta)] \geq 0 \quad (11)$$

- 127 7. **Deterministic Boundaries:** For any $\theta \in \mathbb{R}^k$, we have

$$\lim_{\beta \rightarrow \infty} \psi(\beta \theta) = \mathbb{E}_{z \sim q_{\text{dep}}} \left[\alpha_z \text{sign} \left(\alpha_z^\top \frac{\theta}{\|\theta\|} \right) \right] \quad (12)$$

128 This formulation reveals the relationship between scaling, individual welfare, and social welfare. For
 129 each individual n , monotonic scaling implies

$$\frac{\partial W_n(\beta \theta)}{\partial \beta} = \theta_n^\top \frac{\partial \psi(\beta \theta)}{\partial \beta} = \mathbb{E}_{z \sim q_{\text{dep}}} [2(\alpha_z^\top \theta_n)(\alpha_z^\top \theta) v(\beta \alpha_z^\top \theta)] \geq 0 \quad (13)$$

130 Stated intuitively, the individual-level welfare impact of scaling depends on whether their utility
 131 covaries with the model's utility across deployment vectors (weighted by current model uncertainty).
 132 The aggregate welfare impact of scaling is a weighted sum of such terms. Some may be positive
 133 (scaling helps) and some may be negative (scaling hurts). This induces a general tradeoff between
 134 maximizing utilitarian welfare and either maintaining a more uniform distribution of benefits (and
 135 harms), or limiting the worst-case impact a model will have on any one individual or group.

136 4.1 Alignment as a Linear Social Choice Problem

137 The preceding result shows that we can re-write the AI alignment problem as the following linear
 138 optimization problem.

139 **Definition 4** (Alignment As Linear Optimization). Agents $n = 1, \dots, N$ each have a private
 140 preference parameter vector $\theta_n \in \mathbb{R}^k$. Each agent's individual welfare is a linear function $W_n(\psi) =$
 141 $\theta_n^\top \psi$. For any exogenously given set of social welfare weights $w \in \Delta_N$, the social welfare function
 142 is a linear function $W(\psi) = \sum_{n=1}^N w_n \theta_n^\top \psi$. The social planner wishes to solve

$$\psi^* \in \arg \max_{\psi \in \Psi} \sum_{n=1}^N w_n \theta_n^\top \psi \quad (14)$$

143 where the production possibilities set $\bar{\Psi}$ is a convex, closed, and bounded set that is symmetric about
 144 the origin.

¹From an economic perspective, ψ is analogous to the quantities delivered and θ_n encodes an agent's marginal utility for each.

145 Note that here we are maximizing over the closure $\bar{\Psi}$, which reflects the fact that the deterministic
146 limit is itself a feasible model in practice.

147 This simplification enables us to begin characterizing key properties of what is and is not possible
148 to achieve for alignment while abstracting away from details that don't impact the tradeoff between
149 individual and social welfare, on the one hand, and the strategic and computational challenges of
150 implementing an efficient solution, on the other.

151 *Remark 5.* For any non-empty closed convex set A , the support function for A is

$$h_A(x) = \sup_{a \in A} x^\top a \quad (15)$$

152 The maximum attainable social welfare from solving (14) is $W^*(\theta) = h_{\bar{\Psi}}(\bar{\theta})$.

153 **Proposition 6.** $\psi^* = \lim_{\beta \rightarrow \infty} \psi(\beta \bar{\theta}_w)$

154 5 Suboptimality of Anonymous RLHF

155 This section formalizes how anonymous RLHF induces welfare distortions. We define two notions
156 of *alignment externality*—the welfare impact of an agent's participation in the alignment process—
157 and show that RLHF fails to internalize them. The impact vector framework makes the welfare
158 consequences transparent: every distortion effect reduces to an inner product with the *impact gap*
159 between the utilitarian optimum and the RLHF solution.

160 5.1 Alignment Externalities

161 We define two complementary notions of alignment externality, corresponding to discrete participation
162 and marginal reweighting. The former is analogous to both the VCG pivot from mechanism design
163 and the data counterfactual from machine learning, while the latter is analogous to the influence
164 function from robust statistics.

165 **The Alignment Externality (Participation Pivot)** The alignment externality measures the welfare
166 impact of including an agent's data in the training process, relative to the counterfactual in which they
167 are excluded. This is the alignment analogue of the VCG pivot from mechanism design: it quantifies
168 how much an agent's participation affects others.

169 For an arbitrary estimator of the deployment parameter, let $\hat{\theta}$ denote the parameter obtained from
170 training on the full population and $\hat{\theta}_{-n}$ the parameter obtained when agent n is excluded (and
171 remaining agents' weights are renormalized). The corresponding deployed impacts are $\psi = \psi(\hat{\theta})$
172 and $\psi_{-n} = \psi(\hat{\theta}_{-n})$.

173 **Definition 7** (Alignment Externality). The **alignment externality** of agent n on agent m is

$$\Gamma_{n,m} = W_m(\hat{\theta}) - W_m(\hat{\theta}_{-n}) = \theta_m^\top (\psi - \psi_{-n}) \quad (16)$$

174 The second equality follows from $W_m(\theta) = \theta_m^\top \psi(\theta)$. In impact space, the alignment externality is
175 the inner product of m 's preference with the **impact shift** $\Delta_n = \psi - \psi_{-n}$ caused by n 's participation.

176 The impact shift $\Delta_n \in \mathbb{R}^k$ is a single vector that fully characterizes the welfare consequences of
177 n 's participation: agent m benefits from n 's inclusion whenever $\theta_m^\top \Delta_n > 0$ (their preferences are
178 aligned with the impact shift) and is harmed whenever $\theta_m^\top \Delta_n < 0$. The aggregate externality of n on
179 all other agents is

$$\Gamma_{n,-n} = \sum_{m \neq n} w_m \Gamma_{n,m} = \bar{\theta}_{-n}^\top \Delta_n \quad (17)$$

180 where $\bar{\theta}_{-n} = \frac{1}{1-w_n} \sum_{m \neq n} w_m \theta_m$ is the leave-one-out utilitarian preference. This is positive when
181 the impact shift is aligned with the interests of the remaining population and negative when n 's
182 participation moves the model in a direction the rest of the population dislikes.

The Marginal Alignment Externality (Influence Function) For an RLHF procedure that trains on a query distribution $q_{\text{train}} \in \Delta_{N \times \mathcal{Z}}$, we can also define the *marginal* alignment externality: the first-order welfare effect of infinitesimally upweighting a specific training source (n, z) . This is the alignment analogue of the influence function from robust statistics [see Hammoudeh and Lowd, 2024, for a review].

Definition 8 (Marginal Alignment Externality). For a training source (n, z) , let $q_\varepsilon = (1 - \varepsilon) q_{\text{train}} + \varepsilon \delta_{(n, z)}$. The **marginal alignment externality** of source (n, z) on agent m is

$$\gamma_{n, z}^m = \left. \frac{d}{d\varepsilon} W_m(\hat{\theta}_{q_\varepsilon}) \right|_{\varepsilon=0} = \theta_m^\top \left. \frac{d\psi(\hat{\theta}_{q_\varepsilon})}{d\varepsilon} \right|_{\varepsilon=0} \quad (18)$$

This is the inner product of m 's preference with the **marginal impact shift** from upweighting source (n, z) .

The alignment externality Γ and the marginal externality γ are related: the former is the discrete (finite) welfare change from participation; the latter is the infinitesimal welfare change from reweighting. In impact space, both are inner products of preferences with impact shifts—the alignment externality uses the finite shift $\Delta_n = \psi - \psi_{-n}$, while the marginal externality uses the infinitesimal shift $d\psi/d\varepsilon$. When the alignment procedure is smooth in the training distribution, the alignment externality can be approximated by integrating the marginal externalities, but the two are conceptually distinct: Γ asks “what if agent n had never participated?” while γ asks “what if we gave source (n, z) slightly more weight?”

Lemma 9. Suppose $\hat{\theta}$ is formed by applying a maximum likelihood estimator to training data distributed according to q_{train} . Under regularity conditions on the MLE (stated in the Appendix), the marginal impact shift and the marginal alignment externality satisfy

$$\left. \frac{d\psi(\hat{\theta}_{q_\varepsilon})}{d\varepsilon} \right|_{\varepsilon=0} = \nabla_\theta \psi(\hat{\theta}) \mathcal{J}(\hat{\theta}; q_{\text{train}})^{-1} s_{n, z}(\hat{\theta}) \quad \text{and} \quad \gamma_{n, z}^m = \nabla_\theta W_m(\hat{\theta})^\top \mathcal{J}(\hat{\theta}; q_{\text{train}})^{-1} s_{n, z}(\hat{\theta}) \quad (19)$$

where $\mathcal{J}(\theta; q_{\text{train}}) = -\nabla_\theta^2 \ell(\theta, q_{\text{train}})$ is the population information matrix and $s_{n, z}(\theta) = \mathbb{E}_{c|n, z} [\nabla_\theta \log p_\theta(c | z)]$ is the score contribution of individual n on problem z .

Under the logistic model, the score and welfare gradient take the explicit forms

$$s_{n, z}(\theta) = (\sigma(\alpha_z^\top \theta_n) - \sigma(\alpha_z^\top \theta)) \alpha_z \quad \nabla_\theta W_m(\theta; q_{\text{dep}}) = \mathbb{E}_{z \sim q_{\text{dep}}} [(\alpha_z^\top \theta_m) v(\alpha_z^\top \theta) \alpha_z] \quad (20)$$

5.2 Anonymous RLHF and the Closed-Form Marginal Externality

Anonymous RLHF fits a single parameter $\hat{\theta}^{\text{RLHF}}$ to pooled data, ignoring agent identities at training time:

$$\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} \in \arg \max_{\theta \in \Theta} -\mathbb{E}_{(n, z) \sim q_{\text{train}}} \mathbb{E}_{c|n, z} [-c \log \sigma(\alpha_z^\top \theta) - (1 - c) \log (1 - \sigma(\alpha_z^\top \theta))] \quad (21)$$

Combining the logistic score and information matrix with (19) yields a closed-form expression for the marginal alignment externality under anonymous RLHF

$$\gamma_{n, z}^{\text{RLHF}} = \underbrace{(\sigma(\alpha_z^\top \theta_n) - \sigma(\alpha_z^\top \hat{\theta}^{\text{RLHF}}))}_{\text{preference deviation on query } z} \underbrace{(\alpha_z^\top \eta)}_{\text{welfare sensitivity of query } z} \quad (22)$$

where

$$\eta = \mathbb{E}_{(n, z) \sim q_{\text{train}}} \left[v(\alpha_z^\top \hat{\theta}^{\text{RLHF}}) \alpha_z \alpha_z^\top \right]^{-1} \mathbb{E}_{z \sim q_{\text{dep}}} \left[(\alpha_z^\top \bar{\theta}) v(\alpha_z^\top \hat{\theta}^{\text{RLHF}}) \alpha_z \right] \quad (23)$$

is independent of (n, z) and acts as a sufficient statistic for the welfare implications of parameter changes. Anonymous RLHF distorts welfare along directions that are simultaneously (i) systematically misrepresented in the training data and (ii) socially important at deployment.

5.3 RLHF Stationarity vs. Utilitarian Efficiency

The central result of this section is that RLHF’s first-order condition and utilitarian efficiency impose qualitatively different requirements on the marginal externalities. RLHF ensures only that the *social welfare effect* of marginal perturbations cancels on average across training sources; utilitarian efficiency requires that it vanishes for *every* training source individually.

Definition 10. The **social welfare effect** of upweighting training source (n, z) is the marginal change in utilitarian welfare:

$$\gamma_{n,z}^W = \sum_{m=1}^N w_m \gamma_{n,z}^m = \bar{\theta}_w^\top \frac{d\psi(\hat{\theta}_{q_\varepsilon})}{d\varepsilon} \Big|_{\varepsilon=0} \quad (24)$$

This is the welfare-weighted sum of marginal externalities across all agents, or equivalently, the inner product of the utilitarian preference with the marginal impact shift.

Proposition 11 (RLHF Stationarity vs. Utilitarian Efficiency). *Let $\psi^{\text{RLHF}} = \psi(\hat{\theta}^{\text{RLHF}})$ be the impact deployed by anonymous RLHF and $\psi^* = \arg \max_{\psi \in \bar{\Psi}} \bar{\theta}_w^\top \psi$ the utilitarian-optimal impact.*

1. **RLHF stationarity (average zero):** At ψ^{RLHF} , the training-weighted average social welfare effect vanishes:

$$\mathbb{E}_{(n,z) \sim q_{\text{train}}} [\gamma_{n,z}^W] = 0 \quad (25)$$

Individual sources may have $\gamma_{n,z}^W > 0$ (upweighting would improve welfare) or $\gamma_{n,z}^W < 0$ (upweighting would reduce welfare), but these cancel on average under q_{train} .

2. **Utilitarian efficiency (pointwise zero):** At ψ^* , the social welfare effect vanishes for every training source individually:

$$\gamma_{n,z}^W = 0 \quad \forall (n, z) \in \text{supp}(q_{\text{train}}) \quad (26)$$

No marginal reweighting of any training source can improve social welfare.

3. **The impact gap:** Define $\Delta\psi = \psi^* - \psi^{\text{RLHF}}$. The welfare loss from RLHF is

$$W(\bar{\theta}_w) - W(\hat{\theta}^{\text{RLHF}}) = \bar{\theta}_w^\top \Delta\psi \geq 0 \quad (27)$$

with equality only when $\psi^{\text{RLHF}} = \psi^*$.

4. **Distributional impact:** Agent n ’s welfare change from deploying ψ^{RLHF} instead of ψ^* is

$$W_n(\hat{\theta}^{\text{RLHF}}) - W_n(\bar{\theta}_w) = -\theta_n^\top \Delta\psi \quad (28)$$

Agents whose preferences are aligned with the impact gap ($\theta_n^\top \Delta\psi > 0$) are harmed by RLHF’s distortion; agents anti-aligned with $\Delta\psi$ benefit.

The welfare consequences of RLHF distortion are fully characterized by the single vector $\Delta\psi \in \mathbb{R}^k$: the aggregate welfare loss is $\bar{\theta}_w^\top \Delta\psi$, and the per-agent redistribution is $(\theta_n^\top \Delta\psi)_{n=1}^N$.

Remark 12 (Query-level decomposition). The impact gap decomposes across deployment queries as

$$\bar{\theta}_w^\top \Delta\psi = \mathbb{E}_{z \sim q_{\text{dep}}} \left[(\bar{\theta}_w^\top \alpha_z) \left(\text{sign}(\alpha_z^\top \bar{\theta}_w) - 2(\sigma(\alpha_z^\top \hat{\theta}^{\text{RLHF}}) - \frac{1}{2}) \right) \right] \quad (29)$$

The distortion on each query z is the difference between the deterministic utilitarian decision and the probabilistic RLHF decision, weighted by the welfare importance of the query.

5.4 Recoverability Under Anonymous RLHF

The foregoing analysis establishes that RLHF’s welfare loss can be recast as a failure to internalize alignment externalities. A natural question is whether reweighting the training distribution can close the gap. The answer depends on whether the reweighting conditions on individual identities.

Let $S = \text{span}\{\alpha_z : z \in \text{supp}(q_{\text{train}})\}$ and P_S denote projection onto S .

248 **Definition 13.** A parameter $\theta \in \mathbb{R}^k$ is **individually mix-recoverable** if there exists, for each query
 249 $z \in Z$, a probability vector $\pi(\cdot | z) \in \Delta_N$ such that

$$\sigma(\alpha_z^\top \theta) = \sum_{n=1}^N \pi(n | z) \sigma(\alpha_z^\top \theta_n) \quad \text{for every } z \in Z \quad (30)$$

250 Let Θ^{indiv} denote the set of all such parameters. A parameter is **anonymously mix-recoverable** if
 251 $\pi(n | z) = \frac{1}{N}$ for every n . Denote the set of such parameters Θ^{anon} .

252 **Proposition 14.** 1. A parameter $\theta \in \Theta^{\text{indiv}}$ if and only if $\alpha_z^\top \theta \in [\min_n \alpha_z^\top \theta_n, \max_n \alpha_z^\top \theta_n]$
 253 for every $z \in Z$.

254 2. $P_S C \subseteq P_S \Theta^{\text{indiv}}$: the convex hull of individual preferences is contained in the individually
 255 recoverable set.

256 Thus, by reweighting queries on an individual-by-individual basis, RLHF can reach any utilitarian-
 257 efficient parameter. However, individual-level reweighting is generally necessary:

258 **Proposition 15.** If $S \neq \{0\}$, then $P_S \Theta^{\text{anon}}$ is either empty or a singleton.

259 Anonymous RLHF is pinned to at most one parameter in the identifiable subspace, regardless of how
 260 the query distribution is chosen.

261 **Corollary 16** (Impact-space consequence). Let $\Psi^{\text{indiv}} = \{\psi(\theta) : \theta \in \Theta^{\text{indiv}}\}$ and $\Psi^{\text{anon}} = \{\psi(\theta) :$
 262 $\theta \in \Theta^{\text{anon}}\}$. Then $\psi(\theta_w) \in \Psi^{\text{indiv}}$ for every $w \in \Delta_N$, but Ψ^{anon} is at most a singleton. The welfare
 263 gap from the anonymity constraint is $\theta_w^\top (\psi^* - \psi^{\text{anon}})$.

264 Letting $\mu_z = \mathbb{E}_{n \sim q_{\text{train}}(n|z)}[\sigma(\alpha_z^\top \theta_n)]$ denote the population-level choice frequency for z , any-
 265 mous RLHF effectively computes the M -projection of the choice data onto the logistic model
 266 family:

$$\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} \in \arg \min_{\theta \in \mathbb{R}^k} \mathbb{E}_{(z,n) \sim q_{\text{train}}} [\text{D}_{\text{KL}}(\text{Bern}(\mu_z) \parallel \text{Bern}(\sigma(\alpha_z^\top \theta)))] \quad (31)$$

267 **Proposition 17.** $P_S \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} \in P_S \Theta^{\text{indiv}}$ if $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ achieves zero population (asymptotic) training
 268 loss.

269 5.5 Distortion Bounds

270 The impact gap $\bar{\theta}_w^\top \Delta \psi$ can be bounded in terms of the maximum per-query preference deviation
 271 $d_z = \max_n |\alpha_z^\top (\theta_n - \theta)|$.

Proposition 18.

$$\bar{\theta}_w^\top \Delta \psi \leq 2 \|\bar{\theta}\|_{\mathcal{J}_{\text{dep}}(\bar{\theta})} \sqrt{\mathbb{E}_{z \sim q_{\text{dep}}} [v(\alpha_z^\top \bar{\theta}) d_z^2]} + \frac{1}{4} \mathbb{E}_{z \sim q_{\text{dep}}} [|\alpha_z^\top \bar{\theta}| d_z^2] \quad (32)$$

272 The first term is first-order in the preference deviation (scaled by the Fisher information geometry)
 273 and the second is second-order (scaled by welfare importance). Both vanish as $d_z \rightarrow 0$ (preferences
 274 become homogeneous).

275 6 Strategyproof Mechanisms for Alignment

276 **Definition 19.** A **mechanism** is a function $\mu : \mathbb{R}^{N \times k} \rightarrow \Delta(\bar{\Psi})$ that maps a report profile $\hat{\theta}_n$ to a
 277 probability distribution over the space of impact vectors $\bar{\Psi}$.

278 **Definition 20.** A mechanism is **strategy proof** if, for all preference profiles $\theta \in \mathbb{R}^{k \times N}$ and $n \in [N]$,
 279 all θ'_n , and all $\psi \in \bar{\Psi}$,

$$\mathbb{E}[\theta_n^\top \mu(\theta)] \geq \mathbb{E}[\theta_n^\top \mu(\theta'_n, \theta_{-n})] \quad (33)$$

280 where (θ'_n, θ_{-n}) denotes the profile obtained by replacing agent n 's true preferences with θ'_n .

281 **Remark 21.** If a mechanism is strategy proof, then each agent's payoffs are invariant to the scale of
 282 their reports, i.e., for all n ,

$$\mathbb{E}[\theta_n^\top \mu(\beta \theta_n, \theta_{-n})] = \mathbb{E}[\theta_n^\top \mu(\theta_n, \theta_{-n})] \quad (34)$$

283 for all $\beta > 0$

Thus, strategy-proofness requires that a mechanism normalize the length of preferences, in effect projecting them onto the unit ball. Intuitively, if this were not the case, exaggerating the magnitude of one’s reports could yield a profitable deviation. This immediately implies that no strategy-proof mechanism can elicit *absolute* preference intensities, only *relative* preference intensities.²

Definition 22. A mechanism μ is **unanimous** if, for any profile θ such that $\theta_n = \bar{\theta}$ for all $n \in [N]$, $\mu(\theta) = \delta(\arg \max_{\psi \in \Psi} \bar{\theta}^\top \psi)$

Definition 23. A mechanism μ is called a **random dictatorship** with weights $\lambda \in \Delta_N$ if

$$\mu(\theta) = \sum_{n=1}^N \lambda_n \delta \left(\arg \max_{\psi \in \Psi} \theta_n^\top \psi \right) \quad (35)$$

Proposition 24. Assume $\dim(\bar{\Psi}) \geq 2$ and that the query space is rich enough that $\bar{\Psi}$ is strictly convex. Under these conditions, a mechanism for the AI social choice problem is strategy proof and unanimous if and only if it is a random dictatorship for some weights $\lambda \in \Delta_N$.

The proof shows that the AI social choice problem satisfies the conditions of Theorem 1 from Dutta et al. [2002].³

The typical interpretation of a random dictator is that an agent is randomly chosen, then their preferences are implemented. Note, however, that by linearity we can solve

$$\psi_n^* = \arg \max_{\psi \in \Psi} \theta_n^\top \psi \quad (36)$$

and then solve for

$$\theta_\lambda^{\text{sp}} = \psi^{-1} \left(\sum_{n=1}^N \lambda_n \psi_n^* \right) \quad (37)$$

This impact vector linearly interpolate between the agent-optimal impact vectors. Even though it is a single, static parameterization, the resulting model that it yields, $\sigma(\alpha_z^\top \theta_\lambda^{\text{sp}})$ will generate stochastic responses which, *on average* will reproduce the same distribution of responses as a two-level sampling random dictatorship. This is a remarkable fact, and follows from the fact that Ψ is convex and ψ is invertible.

Next, it is natural to ask how much strategyproofness constraints what the planner can achieve. First, note that we can select a set of weights λ to achieve any $\psi \in \Psi^{\text{sp}} = \text{Hull}(\{\psi_n^*\}_{n=1}^N)$. The question of how much smaller Ψ^{sp} is than Ψ depends on how much diversity there is in the distribution of agent preferences. Intuitively, each ψ_n^* is a projection of θ_n onto Ψ . If agent preferences are “dense” in the sense that they spread over all directions, then their convex hull Ψ^{sp} will approximate Ψ . Distortion will be high when social welfare maximization requires that the model select a “compromise” impact that is not itself similar to any individual agent’s preferences.

Given that utilitarian social welfare is linear in the impact vector and Ψ^{sp} is a convex hull, it will generally be uniquely maximized by placing all weight on a single agent (or a small subset of agents) where $\lambda_n > 0$ only if

$$n \in \arg \max_{m \in [N]} \bar{\theta}^\top \psi_m^* \quad (38)$$

Another potential solution is to project the optimal impact vector onto the strategy-proof set,

$$\arg \max_{\psi \in \Psi^{\text{sp}}} \psi^\top \psi^* \quad (39)$$

Yet another potential solution is to use the welfare weights $\lambda = w$, or simply the egalitarian $\frac{1}{N}$.

²Note that the welfare weights could be adjusted to offset this normalization by setting $\tilde{w}_n = w_n / \|\bar{\theta}_n\|$ and $\tilde{\theta}_n = \bar{\theta}_n / \|\bar{\theta}_n\|$. Thus, this projection required by strategy-proofness is formally equivalent to a re-weighting of welfare. One interpretation of noramalization is that it is the manifestation of equal utility weighting in this linear-product welfare space: utility is only defined up to a scale factor; once translated into the linear product space via Bradley-Terry-Luce, the same challenge of inter-personal comparisons arises.

³Stochasticity is more natural here than in other contexts because the compromises we were looking at in preference space were, implicitly, controlling social welfare. In a sense, the stochastic nature of LLM outputs makes “stochastic dictatorship” much more natural; stochasticity is a form of compromise. It only operates when, as we have assumed, choices can be separated (and utility is linear in them, at least in expectation). It assumes, as we have, that a model generates utility on a choice-by-choice basis, rather than as a “gestalt.”

316 **Definition 25.** The **distortion** of a mechanism μ is the ratio of the expected social welfare it achieves
 317 relative to the maximal utilitarian social welfare.

$$\text{Dist}(\mu) = \frac{\mathbb{E}[W(\mu(\theta))]}{\mathbb{E}[W^*(\theta)]} \quad (40)$$

318 The distortion can be re-written in a form that reveals its connection to voting:

Proposition 26.

$$\text{Dist}(\mu) = \frac{\bar{\theta}^\top \psi_\lambda^{\text{sp}}}{\bar{\theta}^\top \psi^*} = \frac{\mathbb{E}_{z \sim q_{\text{dep}}} \left[(\alpha_z^\top \bar{\theta}) \sum_{n=1}^N \lambda_n \text{sign}(\alpha_z^\top \theta_n) \right]}{\mathbb{E}_{z \sim q_{\text{dep}}} \left[(\alpha_z^\top \bar{\theta}) \text{sign}(\sum_{n=1}^N w_n \alpha_z^\top \theta_n) \right]} \quad (41)$$

319 On a given query, the utilitarian-optimal mechanism (denominator) effectively aggregates votes of
 320 welfare-weighted realized utility. The strategy-proof mechanism (numerator), by contrast, effectively
 321 aggregates votes by mechanism weights alone.

322 In this formulation, the externalities are particularly straightforward. When n participates in the
 323 strategy-proof mechanism, the utility of agent m shifts by

$$\Gamma_{n,m} = \lambda_n \theta_m^\top (\psi_n^* - \psi_{\lambda, -n}^{\text{sp}}) \quad (42)$$

324 where $\psi_{\lambda, -n}^{\text{sp}} = \frac{1}{1-\lambda_n} \sum_{i \neq n} \lambda_i \psi_i^*$ is the strategy-proof impact vector without agent n 's report.

325 **[Michelle: Inverted definition below to align with the literature]**

326 **Definition 27** (Distortion). For a fixed preference profile $\theta \in \mathbb{R}^{N \times k}$ with $W^*(\theta) > 0$, define the
 327 **distortion** of a mechanism μ at θ by

$$\text{Dist}(\mu; \theta) = \begin{cases} \frac{W^*(\theta)}{\mathbb{E}[W(\mu(\theta))]}, & \text{if } \mathbb{E}[W(\mu(\theta))] > 0, \\ +\infty, & \text{if } \mathbb{E}[W(\mu(\theta))] \leq 0. \end{cases} \quad (43)$$

328 The **worst-case distortion** of μ is

$$\text{Dist}(\mu) = \sup_{\theta : W^*(\theta) > 0} \text{Dist}(\mu; \theta). \quad (44)$$

329 **Proposition 28** (SP mechanism has unbounded distortion.). Suppose $\bar{\Psi} = \{\psi \in \mathbb{R}^2 : \|\psi\|_2 \leq 1\}$,
 330 and let $w_1 = w_2 = \lambda_1 = \lambda_2 = \frac{1}{2}$. Let μ^{eq} denote the corresponding egalitarian random dictatorship.
 331 Then μ^{eq} has unbounded distortion:

$$\text{Dist}(\mu^{\text{eq}}) = +\infty. \quad (45)$$

332 In particular, for the family of preference profiles

$$\theta_1 = (2, 0), \quad \theta_2 = (-1, \varepsilon), \quad \varepsilon > 0, \quad (46)$$

333 we have

$$\text{Dist}(\mu^{\text{eq}}; \theta) = \frac{2(1 + \varepsilon^2)}{\varepsilon^2 + \sqrt{1 + \varepsilon^2} - 1}, \quad (47)$$

334 and therefore

$$\lim_{\varepsilon \rightarrow 0} \text{Dist}(\mu^{\text{eq}}; \theta) = +\infty. \quad (48)$$

335 *Proof.* For the Euclidean unit ball, the agent-optimal impact is

$$\psi_n^* = \arg \max_{\psi \in \bar{\Psi}} \theta_n^\top \psi = \frac{\theta_n}{\|\theta_n\|_2}. \quad (49)$$

336 Hence

$$\psi_1^* = (1, 0), \quad \psi_2^* = \frac{(-1, \varepsilon)}{\sqrt{1 + \varepsilon^2}}. \quad (50)$$

337 The egalitarian random dictatorship therefore implements

$$\psi_{\text{eq}}^{\text{sp}} = \frac{1}{2}\psi_1^* + \frac{1}{2}\psi_2^*. \quad (51)$$

338 The utilitarian preference vector is

$$\bar{\theta} = \frac{1}{2}(\theta_1 + \theta_2) = \left(\frac{1}{2}, \frac{\varepsilon}{2}\right). \quad (52)$$

339 Since $\bar{\Psi}$ is the unit ball, the utilitarian optimum is

$$\psi^* = \frac{\bar{\theta}}{\|\bar{\theta}\|_2} = \frac{(1, \varepsilon)}{\sqrt{1 + \varepsilon^2}}, \quad (53)$$

340 and therefore

$$W^*(\theta) = \bar{\theta}^\top \psi^* = \|\bar{\theta}\|_2 = \frac{\sqrt{1 + \varepsilon^2}}{2}. \quad (54)$$

341 The welfare achieved by the egalitarian strategy-proof mechanism is

$$\mathbb{E}[W(\mu^{\text{eq}}(\theta))] = \bar{\theta}^\top \psi_{\text{eq}}^{\text{sp}} = \frac{\varepsilon^2 + \sqrt{1 + \varepsilon^2} - 1}{4\sqrt{1 + \varepsilon^2}}. \quad (55)$$

342 Consequently,

$$\text{Dist}(\mu^{\text{eq}}; \theta) = \frac{W^*(\theta)}{\mathbb{E}[W(\mu^{\text{eq}}(\theta))]} = \frac{2(1 + \varepsilon^2)}{\varepsilon^2 + \sqrt{1 + \varepsilon^2} - 1}. \quad (56)$$

343 As $\varepsilon \rightarrow 0$, the denominator is $O(\varepsilon^2)$, so the ratio diverges to $+\infty$. Therefore the worst-case distortion
344 of μ^{eq} is unbounded. $\square$

345 **Proposition 29** (General unbounded distortion of non-dictatorial random dictatorships). *Assume*
346 *$\dim(\bar{\Psi}) \geq 2$, that $\bar{\Psi} \subset \mathbb{R}^k$ is compact, convex, centrally symmetric, and strictly convex, and that*
347 *the welfare weights satisfy $w_n > 0$ for all $n \in [N]$. Let μ_λ be a random dictatorship with weights*
348 *$\lambda \in \Delta_N$. If λ is not a point mass (i.e., μ_λ is not a dictatorship), then μ_λ has unbounded distortion:*

$$\text{Dist}(\mu_\lambda) = +\infty. \quad (57)$$

349 *In particular, the egalitarian mechanism $\lambda_n = 1/N$ has unbounded distortion for every $N \geq 2$.*

350 *Proof.* Because λ is not a point mass, there exists some agent i such that

$$0 < \lambda_i \leq \frac{1}{2}. \quad (58)$$

351 Pick any boundary point $p \in \partial\bar{\Psi}$. Since $\bar{\Psi}$ is compact and convex, there exists a nonzero supporting
352 normal $u \in \mathbb{R}^k$ such that

$$p \in \arg \max_{\psi \in \bar{\Psi}} u^\top \psi. \quad (59)$$

353 By strict convexity, this maximizer is unique, so

$$p = \arg \max_{\psi \in \bar{\Psi}} u^\top \psi. \quad (60)$$

354 Moreover, since $\bar{\Psi}$ is centrally symmetric and $p \neq 0$, we have

$$h_{\bar{\Psi}}(u) = u^\top p > 0. \quad (61)$$

355 Because $\dim(\bar{\Psi}) \geq 2$, the boundary $\partial\bar{\Psi}$ contains a continuous path from p to $-p$. Let $\gamma : [0, 1] \rightarrow \partial\bar{\Psi}$
356 be such a path with $\gamma(0) = p$ and $\gamma(1) = -p$, and define

$$f(t) := u^\top (\lambda_i p + (1 - \lambda_i) \gamma(t)). \quad (62)$$

357 Then

$$f(0) = u^\top p = h_{\bar{\Psi}}(u) > 0 \quad (63)$$

358 and

$$f(1) = u^\top (\lambda_i p - (1 - \lambda_i) p) = (2\lambda_i - 1) u^\top p \leq 0. \quad (64)$$

359 By continuity, there exists $t^* \in [0, 1]$ such that $f(t^*) = 0$. Let

$$q := \gamma(t^*) \in \partial \bar{\Psi}. \quad (65)$$

360 Again by strict convexity, there exists a supporting normal $v \neq 0$ such that

$$q = \arg \max_{\psi \in \bar{\Psi}} v^\top \psi. \quad (66)$$

361 Now define, for each $\varepsilon > 0$, a preference profile θ^ε by

$$\theta_i^\varepsilon = u, \quad \theta_j^\varepsilon = \varepsilon v \quad \text{for all } j \neq i. \quad (67)$$

362 The agent-optimal impacts are therefore

$$\psi_i^* = p, \quad \psi_j^* = q \quad \text{for all } j \neq i. \quad (68)$$

363 Hence the strategy-proof impact implemented by the random dictatorship is

$$\psi_\lambda^{\text{sp}} = \sum_{n=1}^N \lambda_n \psi_n^* = \lambda_i p + (1 - \lambda_i) q, \quad (69)$$

364 which does not depend on ε .

365 The utilitarian preference vector is

$$\bar{\theta}^\varepsilon = \sum_{n=1}^N w_n \theta_n^\varepsilon = w_i u + \varepsilon \left(\sum_{j \neq i} w_j \right) v. \quad (70)$$

366 Therefore the welfare achieved by the mechanism is

$$\mathbb{E}[W(\mu_\lambda(\theta^\varepsilon))] = (\bar{\theta}^\varepsilon)^\top \psi_\lambda^{\text{sp}} \quad (71)$$

$$= w_i u^\top \psi_\lambda^{\text{sp}} + \varepsilon \left(\sum_{j \neq i} w_j \right) v^\top \psi_\lambda^{\text{sp}} \quad (72)$$

$$= \varepsilon \left(\sum_{j \neq i} w_j \right) v^\top \psi_\lambda^{\text{sp}}, \quad (73)$$

367 since $u^\top \psi_\lambda^{\text{sp}} = f(t^*) = 0$. Thus

$$\mathbb{E}[W(\mu_\lambda(\theta^\varepsilon))] = O(\varepsilon). \quad (74)$$

368 If this quantity is nonpositive for some ε , then $\text{Dist}(\mu_\lambda; \theta^\varepsilon) = +\infty$ by definition. Otherwise it is
369 positive and converges to 0 as $\varepsilon \rightarrow 0$.

370 On the other hand, the optimal utilitarian welfare is

$$W^*(\theta^\varepsilon) = h_{\bar{\Psi}}(\bar{\theta}^\varepsilon). \quad (75)$$

371 Since the support function is continuous and $\bar{\theta}^\varepsilon \rightarrow w_i u$ as $\varepsilon \rightarrow 0$, we have

$$W^*(\theta^\varepsilon) \rightarrow h_{\bar{\Psi}}(w_i u) = w_i h_{\bar{\Psi}}(u) > 0. \quad (76)$$

372 Therefore the numerator stays bounded away from zero while the denominator converges to zero (or
373 becomes nonpositive), implying

$$\text{Dist}(\mu_\lambda; \theta^\varepsilon) \rightarrow +\infty. \quad (77)$$

374 Taking the supremum over preference profiles yields

$$\text{Dist}(\mu_\lambda) = +\infty. \quad (78)$$

375 $\square$

7 Alignment Mechanisms with Welfare Guarantees

7.1 Bounding Harm

We may also be concerned with limiting negative welfare impacts on individuals agents or groups of agents.⁴ For any collection of harm bounds $b \in \mathbb{R}^N$,

$$\psi_b^h \in \arg \max_{\psi \in \Psi} \bar{\theta}^\top \psi \quad (79)$$

$$\text{s.t. } \psi^\top \theta_n \geq b_n \quad \forall n \quad (80)$$

For example, we might want

1. $b_n = -\epsilon$ for all n (constant limited harm)
2. $b_n = \theta_n^\top \psi_w^{\text{sp}}$ (better than the strategy-proof mechanism)
3. $b_n = \nu_n \theta_n^\top \psi_n^*$ for $\nu \in \Delta_N$ (a fraction of the agent-optimal impact)

Conjecture 30.

1. If ψ_b^h exists, it is unique. Moreover, ψ_b^h exists for each of the three examples above.
2. If η_n is the shadow price of the constraint for agent n and we define

$$\theta_b^h = \bar{\theta}_w + \sum_{n=1}^N \eta_n (\theta_n - \bar{\theta}_w) \quad (81)$$

then the optimal solution takes the form

$$\psi_b^h = \arg \max_{\psi \in \Psi} \theta_b^{h\top} \psi \quad (82)$$

By complementary slackness, $\eta_n (\theta_n^\top \psi_b^h - b_n) = 0$ for all n .

7.2 Public Spirit

Perhaps a more flexible possibility is that each person is willing to give up $\nu \geq 0$ units of personal utility per unit of social utility generated for everyone else [Flanigan et al., 2023]. Let ψ_0 be a “reference” impact vector. We can solve

$$\begin{aligned} \psi_\nu^{\text{ps}} &\in \arg \max_{\psi \in \Psi} \bar{\theta}^\top \psi \\ \text{s.t. } (\nu \bar{\theta} + \theta_n)^\top \psi &\geq (\nu \bar{\theta} + \theta_n)^\top \psi_0 \quad \forall n \end{aligned} \quad (83)$$

If $\nu = 0$, then each individual will only accept moves that increase their personal utility: the planner is restricted to Pareto-improving moves. As $\nu \rightarrow \infty$, the planner can impose arbitrarily large harm on any individual, and the problem becomes strict maximization, such that $\lim_{\nu \rightarrow \infty} \psi_\nu^{\text{ps}} = \psi^*$. In the middle, the planner can impose some harm, but not arbitrarily much.

Conjecture 31. If we denote $\eta \in \mathbb{R}_{\geq 0}^N$ to be the shadow prices of the constraints and

$$\theta_\nu^{\text{ps}} = \bar{\theta} + \sum_{n=1}^N \eta_n (\nu \bar{\theta} + \theta_n) \quad (84)$$

then the optimal solution is the projection of θ_ν^{ps} onto Ψ :

$$\psi_\nu^{\text{ps}} \in \arg \max_{\psi \in \Psi} \theta_\nu^{\text{ps}\top} \psi \quad (85)$$

8 Empirical Evaluation

9 Conclusion

⁴We might further subset by deployment contexts. It would be worth circling back to this as an extension.

Proofs

Proof of Lemma 1. The logistic choice function $2(\sigma(\beta \alpha_z^\top \bar{\theta}) - \frac{1}{2})$ converges to an indicator $2(\mathbb{I}(\alpha_z^\top \bar{\theta} \geq 0) - \frac{1}{2})$ as the inverse temperature goes to infinity. Hence

$$\lim_{\beta \rightarrow \infty} W(\beta \bar{\theta}) = \mathbb{E}_{z \sim q_{\text{dep}}} [|\alpha_z^\top \bar{\theta}|] \quad (86)$$

Next, note that for any α_z , we have $2 \alpha_z^\top \bar{\theta} (\sigma(\alpha_z^\top \bar{\theta}) - \frac{1}{2}) \leq |\alpha_z^\top \bar{\theta}|$. Taking an expectation over both sides yields

$$W(\theta) \leq \mathbb{E}_{z \sim q_{\text{dep}}} [\alpha_z^\top \bar{\theta} \sigma(\alpha_z^\top \bar{\theta})] \leq \mathbb{E}_{z \sim q_{\text{dep}}} [|\alpha_z^\top \bar{\theta}|] \quad (87)$$

for all θ , and the result follows. $\square$

Proof of Theorem 3.

1. Define

$$\Phi_{\text{dep}}(\theta) = \mathbb{E}_{z \sim q_{\text{dep}}} \left[2 \left(\log(1 + \exp(\alpha_z^\top \theta)) - \frac{\alpha_z^\top \theta}{2} \right) \right] \quad (88)$$

First, note that

$$\nabla_{\text{dep}} \Phi_{\text{dep}}(\theta) = \psi(\theta) \quad (89)$$

where ∇_{dep} is the gradient operator restricted to S_{dep} . The restriction is without loss valid because $\alpha_z \in S_{\text{dep}}$ for all $z \in \text{supp}(q_{\text{dep}})$ by construction. Second, note that

$$\nabla_{\text{dep}}^2 \Phi_{\text{dep}}(\theta) = \mathbb{E}_{z \sim q_{\text{dep}}} [2 v(\alpha_z^\top \theta) \alpha_z \alpha_z^\top] \quad (90)$$

which is positive-definite over S_{dep} . Therefore, Φ_{dep} is a strictly convex function on S_{dep} . Next, the Legendre transform of Φ_{dep} as

$$\tilde{\Phi}_{\text{dep}}(y) = \sup_{\theta \in S_{\text{dep}}} \{y^\top \theta - \Phi_{\text{dep}}(\theta)\} \quad (91)$$

Further, $\tilde{\Phi}_{\text{dep}}$ is C^∞ . By Theorem 26.5 of Rockafellar (1970), the gradient map $\psi = \nabla \Phi_{\text{dep}}$ is a C^∞ diffeomorphism from S_{dep} onto $\Psi \cap S_{\text{dep}}$ and $\Psi = \text{int}(\text{dom}(\tilde{\Phi}_{\text{dep}}))$ is a convex set.

2. Note that $\sigma(0) = \frac{1}{2}$. By direct calculation,

$$\psi(0) = \mathbb{E}_{z \sim q_{\text{dep}}} \left[2 \alpha_z \left(\sigma(0) - \frac{1}{2} \right) \right] = 0 \quad (92)$$

3. Note that $\sigma(-x) = 1 - \sigma(x)$. Therefore,

$$\psi(-\theta) = \mathbb{E}_{z \sim q_{\text{dep}}} \left[2 \alpha_z \left(\sigma(-\alpha_z^\top \theta) - \frac{1}{2} \right) \right] = \mathbb{E}_{z \sim q_{\text{dep}}} \left[2 \alpha_z \left(1 - \sigma(\alpha_z^\top \theta) - \frac{1}{2} \right) \right] = -\psi(\theta) \quad (93)$$

4. The signoid is bounded between 0 and 1. Therefore, $\sigma(\alpha_z^\top \theta) \in [0, 1]$ for all $\alpha_z, \theta \in \mathbb{R}^k$. Therefore

$$\sup_{\theta \in \mathbb{R}^k} \|\psi(\theta)\| = \sup_{\theta \in \mathbb{R}^k} \left\| \mathbb{E}_{z \sim q_{\text{dep}}} \left[2 \alpha_z \left(\sigma(\alpha_z^\top \theta) - \frac{1}{2} \right) \right] \right\| \leq \mathbb{E}_{z \sim q_{\text{dep}}} [2 \|\alpha_z\|] = 2 \mathbb{E}_{z \sim q_{\text{dep}}} [\|\alpha_z\|] \quad (94)$$

5. For any fixed $\alpha_z \in \mathbb{R}^k$, $2 \alpha_z (\sigma(\alpha_z^\top \theta) - \frac{1}{2})$ is C^∞ in θ . The expectation over C^∞ functions is C^∞ . Therefore, ψ is C^∞ . [Need to establish the inverse.]

6. The result follows from applying differentiaion under the integral and the chain rule.

7. First, note that for any x , $\lim_{\beta \rightarrow \infty} \sigma(\beta x) = \text{sign}(x)$. Second, we can pass the limit inside the expectation over deployment vectors by dominated convergence theorem since $\sigma(\alpha_z^\top \theta) \in [0, 1]$ for all $\alpha_z, \theta \in \mathbb{R}^k$. The result follows.

427

□

428 *Proof of Proposition 6.* The result follows from the definition of ψ and Lemma 1. □429 *Proof of Lemma 9.* First, apply the chain rule

$$\left. \frac{d}{d\varepsilon} W_m(\hat{\theta}(\varepsilon)) \right|_{\varepsilon=0} = \nabla_{\theta} W_m(\hat{\theta}_{q_{\text{train}}})^{\top} \left. \frac{d\hat{\theta}(\varepsilon)}{d\varepsilon} \right|_{\varepsilon=0} \quad (95)$$

430 From there,

$$\left. \frac{d\hat{\theta}(\varepsilon)}{d\varepsilon} \right|_{\varepsilon=0} = \mathcal{J}(\hat{\theta}_{q_{\text{train}}}; q_{\text{train}})^{-1} s_{n,z}(\hat{\theta}_{q_{\text{train}}}) \quad (96)$$

431 is a just standard influence function, and follows from the typical derivation [e.g., Hammoudeh and
432 Lowd, 2024]. Combining these two yields the result. □433 *Proof of Proposition 14.* Part 1: $\Rightarrow$ Per the definition of $\theta \in \Theta^{\text{indiv}}$, there exists a π such that,
434 for all z ,

$$\sigma(\alpha_z^{\top} \theta) = \sum_{n=1}^N \pi(n|z) \sigma(\alpha_z^{\top} \theta_n) \quad (97)$$

435 As a convex combination of scalars, the right-hand side must lie between its extreme
436 points, so that $\sigma(\alpha_z^{\top} \theta) \in [\min_n \sigma(\alpha_z^{\top} \theta_n), \max_n \sigma(\alpha_z^{\top} \theta_n)]$ for all z . But then, by the
437 strict monotonicity of σ , this implies that $\alpha_z^{\top} \theta \in [\min_n \alpha_z^{\top} \theta_n, \max_n \alpha_z^{\top} \theta_n]$ for all z .
438 $\Leftarrow$ Let $z \in Z$ be arbitrary, and suppose that $\sigma(\alpha_z^{\top} \theta) \in [\min_n \sigma(\alpha_z^{\top} \theta_n), \max_n \sigma(\alpha_z^{\top} \theta_n)]$.
439 Let $\underline{n} = \arg \min_n \sigma(\alpha_z^{\top} \theta_n)$ and $\bar{n} = \arg \max_n \sigma(\alpha_z^{\top} \theta_n)$. Then, by the monotonicity
440 of the logistic function, there exists a $\pi(\underline{n}|z) \in [0, 1]$ such that

$$\sigma(\alpha_z^{\top} \theta) = \pi(\underline{n}|z) \sigma(\alpha_z^{\top} \theta_{\underline{n}}) + \pi(\bar{n}|z) \sigma(\alpha_z^{\top} \theta_{\bar{n}}) \quad (98)$$

441 Applying this construction across all $z \in Z$ yields the result.

442 Part 2: $P_S C \subseteq P_S \Theta^{\text{indiv}}$: Fix any $\theta \in C$. By the definition of C , there exists $\theta \in \sum_{n=1}^N \lambda_n \theta_n$
443 for some $\lambda \in \Delta_N$. By the convexity of linear projection, $\alpha_z^{\top} \theta \in [\min_n \alpha_z^{\top} \theta_n, \max_n \alpha_z^{\top} \theta_n]$
444 for all $z \in Z$. But then $\sigma(\alpha_z^{\top} \theta) \in [\min_n \sigma(\alpha_z^{\top} \theta_n), \max_n \sigma(\alpha_z^{\top} \theta_n)]$ for all $z \in Z$. Hence,
445 $\theta \in \Theta^{\text{indiv}}$. □

446

447 *Proof of Proposition 15.* When $\theta \in \Theta^{\text{anon}}$,

$$\alpha_z^{\top} \theta = \sigma^{-1} \left(\frac{1}{N} \sum_{n=1}^N \sigma(\alpha_z^{\top} \theta_n) \right) \quad \forall z \quad (99)$$

448 where the right-hand side is determined entirely by the population $\{\theta_n\}_{n=1}^N$ for each z . If there exists
449 no θ satisfying this system of linear constraints, then $\Theta^{\text{anon}} = \emptyset$ and hence $P_S \Theta^{\text{anon}}$ is empty.

450 Otherwise, suppose $\theta, \theta' \in \Theta^{\text{anon}}$. They both satisfy the system of linear constraints, so for every z ,
451 $\alpha_z^{\top} \theta = \alpha_z^{\top} \theta'$ and hence $\alpha_z^{\top} (\theta - \theta') = 0$ for z . This implies $\theta - \theta'$ is orthogonal to $\text{span}\{\alpha_z : z\} = S$.
452 Thus all elements of Θ^{anon} have the same projection onto S . Therefore $P_S \Theta^{\text{anon}}$ is at most a
453 singleton. □

454 *Proof of Proposition ??.* If $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ achieves zero population training loss, then every KL term van-
455 ishes on the support of q_{train} , hence

$$\sigma(\alpha_z^{\top} \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}) = \sum_{n=1}^N q_{\text{train}}(n | z) \sigma(\alpha_z^{\top} \theta_n) \quad \forall z \in \text{supp}(q_{\text{train}}) \quad (100)$$

456 is exactly the mixture-recovery condition on the identifiable queries. $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ is only identifiable in the
457 subspace S . Applying Proposition 14, this implies that $P_S \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} \in P_S \Theta^{\text{mix}}$. □

458 *Proof of Proposition 15.* When $\theta \in \Theta^{\text{anon}}$,

$$\alpha_z^\top \theta = \sigma^{-1} \left(\frac{1}{N} \sum_{n=1}^N \sigma(\alpha_z^\top \theta_n) \right) \quad \forall z \quad (101)$$

459 where the right-hand side is determined entirely by the population $\{\theta_n\}_{n=1}^N$ for each z . If there exists
460 no θ satisfying this system of linear constraints, then $\Theta^{\text{anon}} = \emptyset$ and hence $P_S \Theta^{\text{anon}}$ is empty.

461 Otherwise, suppose $\theta, \theta' \in \Theta^{\text{anon}}$. They both satisfy the system of linear constraints, so for every z ,
462 $\alpha_z^\top \theta = \alpha_z^\top \theta'$ and hence $\alpha_z^\top (\theta - \theta') = 0$ for z . This implies $\theta - \theta'$ is orthogonal to $\text{span}\{\alpha_z : z\} = S$.
463 Thus all elements of Θ^{anon} have the same projection onto S . Therefore $P_S \Theta^{\text{anon}}$ is at most a
464 singleton. $\square$

465 *Proof of Proposition 17.* If $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ achieves zero population training loss, then every KL term van-
466 ishes on the support of q_{train} , hence

$$\sigma(\alpha_z^\top \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}) = \sum_{n=1}^N q_{\text{train}}(n | z) \sigma(\alpha_z^\top \theta_n) \quad \forall z \in \text{supp}(q_{\text{train}}) \quad (102)$$

467 is exactly the mixture-recovery condition on the identifiable queries. $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ is only identifiable in the
468 subspace S . Applying Proposition 14, this implies that $P_S \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} \in P_S \Theta^{\text{mix}}$. $\square$

469 **Lemma 32** (Lemma 3.2 of Dutta et al. [2002]). *Let a binary preference relation $\succeq$ over a non-empty*
470 *convex action space $A \subseteq \mathbb{R}^k$ satisfying the following be given:*

- 471 1. **Single-Peakedness:** *There is a unique a^* such that $a^* \succeq a$ for all $a \in A$.*
- 472 2. **Convexity:** *For all $a, a' \in A$ such that $a' \succeq a$ and $a' \neq a$, for all $\lambda \in [0, 1]$, we have*
473 *$(\lambda a' + (1 - \lambda)a) \succeq a$.*
- 474 3. **Closed Level Sets:** *For all $a \in A$, the sets $\{a' \in A : a' \succeq a\}$ and $\{a' \in A : a \succeq a'\}$ are*
475 *closed.*

476 *Let $\dim(A) \geq 2$ and μ be a stochastic mechanism that maps from report profiles to probability*
477 *distributions over A . Then μ is strategy-proof and unanimous if and only if it is a random dictatorship.*

478 *Proof of Proposition 24.* We show that the linear preferences of the alignment problem satisfy the
479 conditions of Lemma 32 for each agent n when restricted to the closure $\bar{\Psi}$.

- 480 1. **Single-Peakedness:** For each agent n , let $\psi_n^* = \arg \max_{\psi \in \bar{\Psi}} \theta_n^\top \psi$ be the agent-optimal
481 impact vector. By strict convexity of $\bar{\Psi}$, this projection is unique.
- 482 2. **Convexity:** For each agent n , the function $\tilde{W}_n(\psi) = \theta_n^\top \psi$ is linear and therefore has convex
483 upper level sets.
- 484 3. **Closed Level Sets:** Let $\psi \in \bar{\Psi}$ be given. The sets $\{\psi' \in \bar{\Psi} : \theta_n^\top \psi' \geq \theta_n^\top \psi\}$ and
485 $\{\psi' \in \bar{\Psi} : \theta_n^\top \psi' \leq \theta_n^\top \psi\}$ are each an intersection of a closed half-plane with the closed set
486 $\bar{\Psi}$, and are therefore both closed.

487 $\square$

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